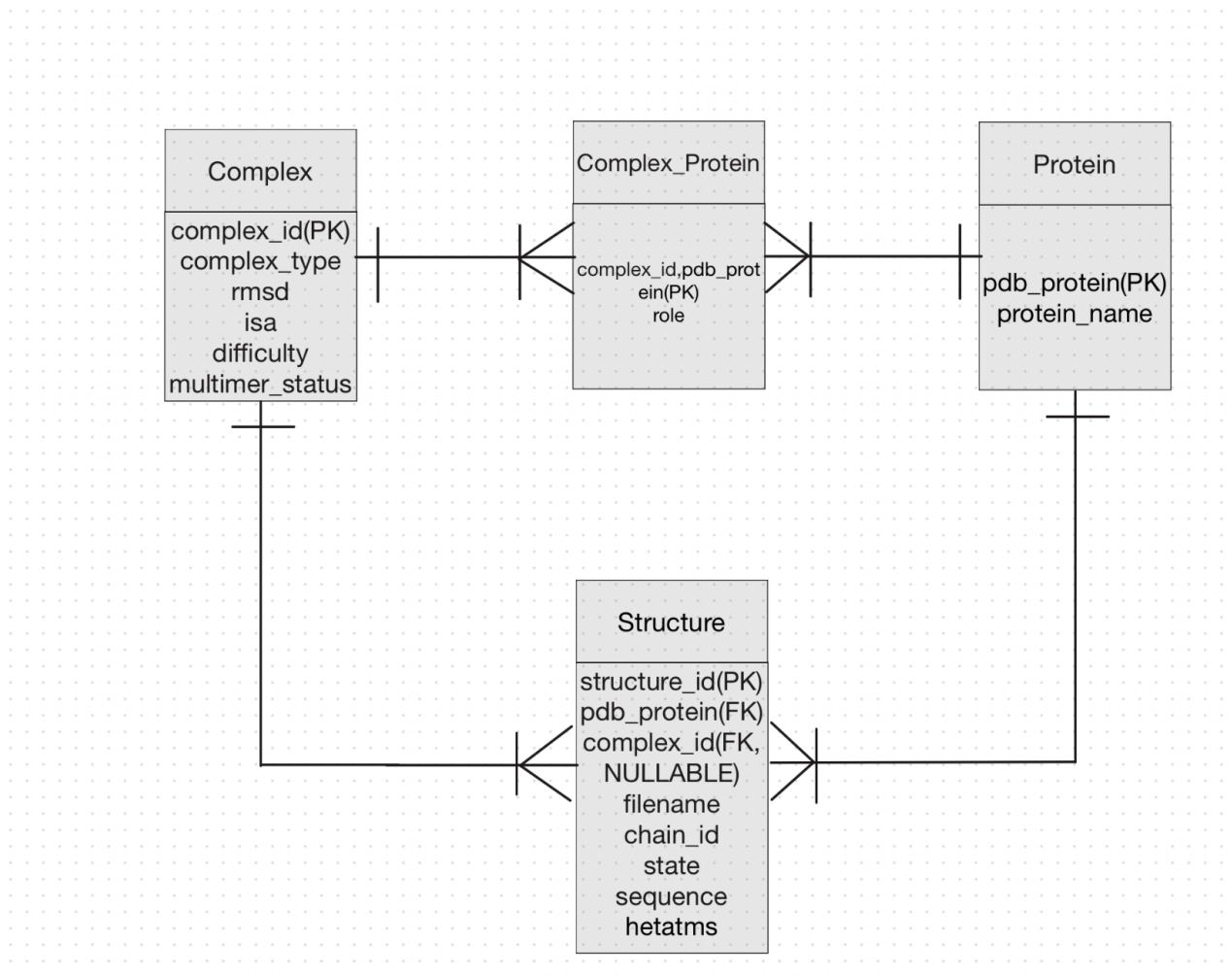


The goal of this database is to allow users to store, query, and analyze structural and functional information about protein complexes.

1. **Design a normalized database schema** for proteins, structures, complexes, and interactions.
2. **Perform ETL tasks** to extract, clean, and load data from PDB files and benchmark metadata.
3. **Write queries** to retrieve meaningful biological information.

Requirements for submission:

1. **Conceptual schema of your database: ER diagram, data dictionary (optional), ontology (optional) (20)**



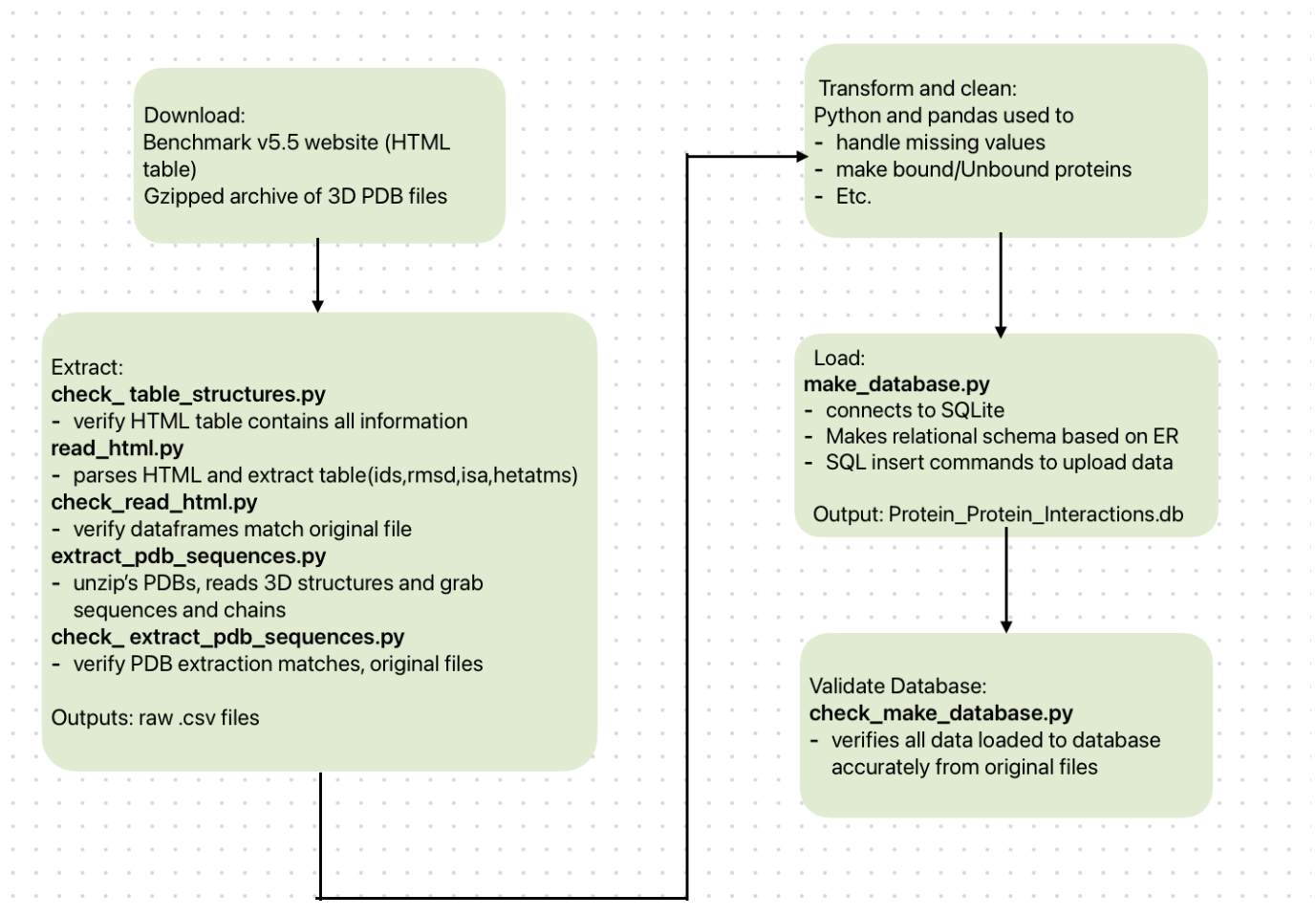
2. **ETL workflow: How did you download, extract and loaded the data onto your database (20)**

I downloaded: [Download cleaned-up PDB files for the benchmark in a gzipped archive](#)

And then used the Protein-Protein Docking Benchmark v5.5 website as a .html file for the table.

To extract and load the data I used python:

1. Check that table from .html contained all information (check_table_structures.py)
2. Script to extract the table from the .html (read_html.py)
3. Check that table (check_read_html.py)
4. Script that unzipped folder of .pdb files one by one, reads the 3D structure, grabs the amino acid sequences and saves it. (extract_pdb_sequences.py)
5. Check the pdb extraction (check_extract_pdb_sequences.py)
6. Script to load the data on the database based on my design (make_database.py)
7. Check the data loaded and compare to the original files (check_make_database.py)



3. Your database submission as .db or .cypher file (30)

I attached Protein_Protein_Interactions.db.

4. Output of example queries mentioned above (30).

1. Query by complex type

- Supported types:
 - **EI** = Enzyme–Inhibitor
 - **ES** = Enzyme–Substrate
 - **ER** = Enzyme with regulatory/accessory chain
 - **AA** = Antibody–Antigen
 - **AS** = Antigen–Single domain Antibody
 - **OG** = Others, G-protein containing
 - **OR** = Others, Receptor containing
 - **OX** = Others, miscellaneous
-

SELECT

complex_type,

COUNT(*) as total_complexes

FROM Complex

GROUP BY complex_type

ORDER BY total_complexes DESC;

	complex_type	total_complexes
1	OX	55
2	AA	55
3	EI	45
4	ER	26
5	OR	24
6	OG	23
7	ES	17
8	AS	12

SELECT complex_id, complex_type, difficulty

FROM Complex

WHERE complex_type = 'AS';

	complex_id	complex_type	difficulty
1	2I25_N:L	AS	rigid
2	4Y7M_A:C	AS	rigid
3	5JMO_D:B	AS	rigid
4	5SV3_C:D	AS	rigid
5	6CWG_B:A	AS	rigid
6	6DBG_C:B	AS	rigid
7	4M3K_B:A	AS	medium
8	4POU_B:A	AS	medium
9	5E5M_B:A	AS	medium
10	5HGG_T:A	AS	rigid
11	5VNW_D:A	AS	rigid
12	6EY6_J:AB	AS	medium

2. Query sequences and structures

- Retrieve:
 - Complex IDs
 - Protein sequences
 - Bound complex structure IDs
 - Unbound protein structure IDs and chain IDs
- **Note:** For this, you will need to extract sequences from PDB files as part of ETL.

SELECT

cp.complex_id,

s.pdb_protein,

s.state,

s.filename as structure_id,

s.chain_id,

s.sequence

FROM Complex_Protein cp

JOIN Structure s ON cp.pdb_protein = s.pdb_protein

WHERE cp.complex_id = '1AHW_AB:C'

ORDER BY s.pdb_protein, s.state DESC;

	complex_id	pdb_prote	state	structure_id	chain_id	sequence
1	1AHW_AB:C	1FGN_LH	Unbound	1AHW_r_u.pdb	H	EIQLQQSGAELVRPGALVKLSCKASGFNIKDYMHWWKQRPEQGLEWIGLIDPENGNTIYDPKFQ GKASI..
2	1AHW_AB:C	1FGN_LH	Unbound	1AHW_r_u.pdb	L	DIKMTQSPSSMYASLGERVTITCKASQDIRKYLNWYQQKPWKSPKTLIYYATSLADGVPSRFSGSGSGQ..
3	1AHW_AB:C	1FGN_LH	Bound	1AHW_r_b.pdb	A	DIKMTQSPSSMYASLGERVTITCKASQDIRKYLNWYQQKPWKSPKTLIYYATSLADGVPSRFSGSGSGQ..
4	1AHW_AB:C	1FGN_LH	Bound	1AHW_r_b.pdb	B	EIQLQQSGAELVRPGALVKLSCKASGFNIKDYMHWWKQRPEQGLEWIGLIDPENGNTIYDPKFQ GKASI..
5	1AHW_AB:C	1TFH_A	Unbound	1AHW_l_u.pdb	A	NTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCFYTTDTECDLTDEIVKDVKQTYLA..
6	1AHW_AB:C	1TFH_A	Bound	1AHW_l_b.pdb	C	TNTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCFYTTDTECDLTDEIVKDVKQTYL..

3. Query by docking difficulty

- Levels: rigid, medium, difficult

SELECT

difficulty,

COUNT(complex_id) AS total_complexes,

ROUND(AVG(rmsd), 2) AS average_rmsd,

MIN(rmsd) AS min_rmsd,

MAX(rmsd) AS max_rmsd

FROM Complex

WHERE difficulty IS NOT NULL

GROUP BY difficulty

ORDER BY average_rmsd;

	difficulty	total_complexes	average_rmsd	min_rmsd	max_rmsd
1	rigid	177	0.84	0.29	1.49
2	medium	45	1.87	1.55	2.2
3	difficult	35	3.48	2.26	8.38

4. Check protein occurrence in multiple complexes

- Identify proteins that participate in more than one complex

```
SELECT
```

```
    pdb_protein,
```

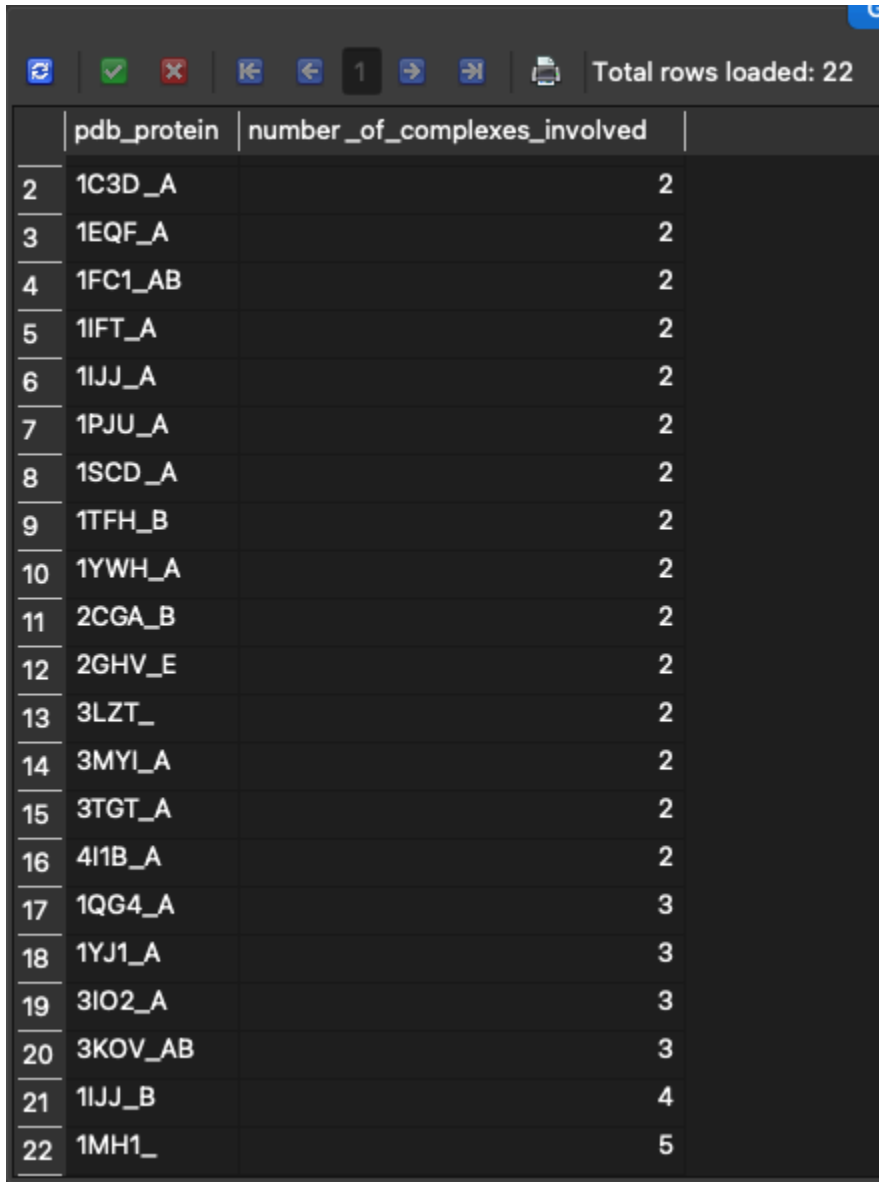
```
    COUNT(complex_id) AS number_of_complexes_involved
```

```
FROM Complex_Protein
```

```
GROUP BY pdb_protein
```

```
HAVING number_of_complexes_involved > 1
```

ORDER BY number_of_complexes_involved ;



The screenshot shows a data table interface with a toolbar at the top containing icons for refresh, check, delete, first, previous, page 1, next, last, and print. The text 'Total rows loaded: 22' is displayed on the right. The table has three columns: an index column, 'pdb_protein', and 'number_of_complexes_involved'. The data is sorted by the third column in ascending order.

	pdb_protein	number_of_complexes_involved
2	1C3D_A	2
3	1EQF_A	2
4	1FC1_AB	2
5	1IFT_A	2
6	1IJJ_A	2
7	1PJU_A	2
8	1SCD_A	2
9	1TFH_B	2
10	1YWH_A	2
11	2CGA_B	2
12	2GHV_E	2
13	3LZT_	2
14	3MYI_A	2
15	3TGT_A	2
16	4I1B_A	2
17	1QG4_A	3
18	1YJ1_A	3
19	3IO2_A	3
20	3KOV_AB	3
21	1IJJ_B	4
22	1MH1_	5

5. Maintain all original dataset features

- RMSD
- HETATMs
- Interface surface area (iSA)

SELECT DISTINCT

c.complex_id,
c.complex_type,

```

c.rmsd,
c.isa,
c.difficulty,
c.multimer_status,
s.filename,
s.hetatms
FROM Complex c
JOIN Structure s ON c.complex_id = s.complex_id
WHERE s.state = 'Bound' ;

```

		Grid view Form view							
		Total rows loaded: 506							
	complex_id	complex_type	rmsd	isa	difficulty	multimer_status	filename	hetatms	
1	2OUL_A:B	EI	0.53	1933	rigid	NULL	2OUL_l_b.pdb		
2	1EXB_ABDC:EGFH	OX	0.62	3558	rigid	Complex Dimer of Tetramers Created	1EXB_r_b.pdb	NDP	
3	2DD8_HL:S	AA	2.19	1710	medium	NULL	2DD8_l_b.pdb		
4	2FD6_HL:U	AA	1.07	1139	rigid	NULL	2FD6_l_b.pdb		
5	1DE4_AB:CF	OX	2.59	2066	difficult	NULL	1DE4_l_b.pdb		
6	1R8S_A:E	OG	3.73	2986	difficult	NULL	1R8S_l_b.pdb		
7	6CWG_B:A	AS	0.76	1151	rigid	NULL	6CWG_l_b.pdb		
8	1QA9_A:B	OX	0.73	1353	rigid	NULL	1QA9_l_b.pdb		
9	3S9D_B:A	OR	1.69	1841	medium	NULL	3S9D_r_b.pdb		
10	1JMO_A:HL	ER	3.21	3461	difficult	NULL	1JMO_l_b.pdb		
11	5CBA_AB:E	AA	1.49	1790	rigid	NULL	5CBA_r_b.pdb		
12	1S78_DC:A	AA	1.13	2175	rigid	NULL	1S78_l_b.pdb		
13	7CEI_A:B	EI	0.7	1384	rigid	NULL	7CEI_r_b.pdb		
14	3RVW_CD:A	AA	0.5	1383	rigid	NULL	3RVW_r_b.pdb		
15	1US7_A:B	ER	1.06	1095	rigid	NULL	1US7_r_b.pdb		
16	2FJU_B:A	OG	1.04	1245	rigid	NULL	2FJU_r_b.pdb		
17	1JK_A:BC	ER	0.68	1648	rigid	NULL	1JK_r_b.pdb		
18	2OOB_A:B	ES	0.85	808	rigid	NULL	2OOB_r_b.pdb		
19	1CGI_E:I	EI	2.02	2053	medium	NULL	1CGI_l_b.pdb		
20	2X9A_D:C	OR	1.33	1571	rigid	NULL	2X9A_l_b.pdb		
21	2J7P_A:D	OX	2.67	3008	difficult	NULL	2J7P_r_b.pdb	GDP	